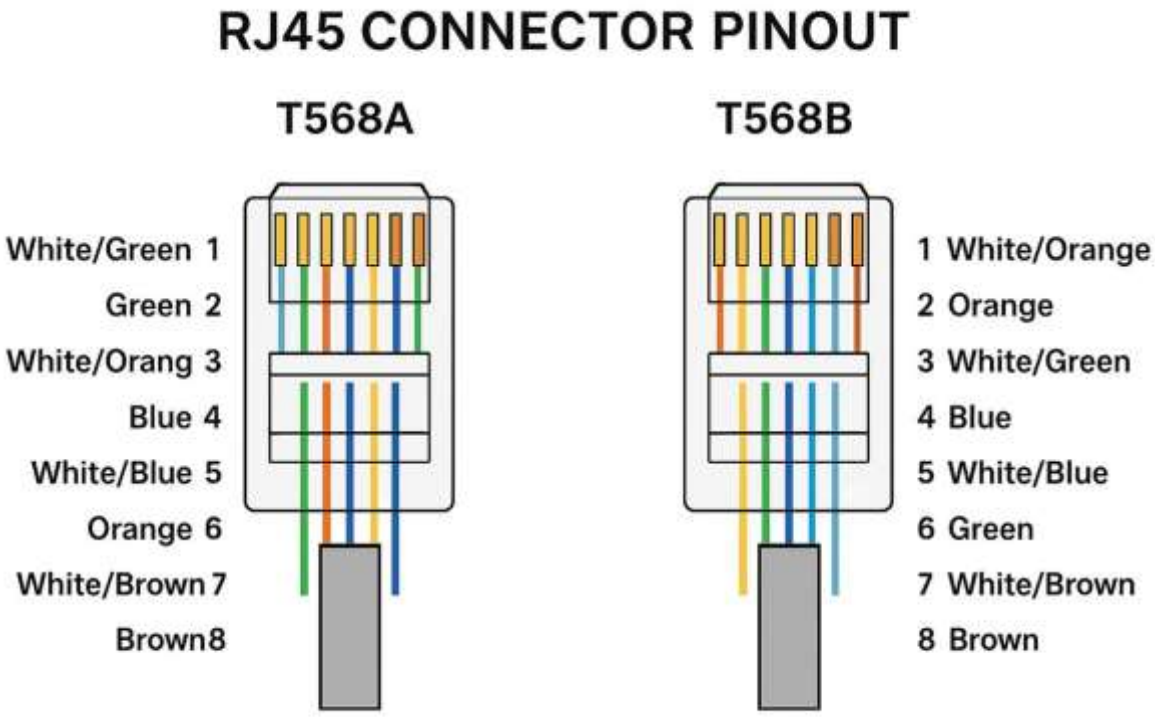
Ethernet Standard And Specification

1.Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

ANS:

RJ-45 is a connector used to connect an Ethernet cable to a computer, router or switch.



Colour Codes

Pin	T568A	T568B
1	White-Green	White-Orange
2	Green	Orange
3	White-Orange	White-Green
4	Blue	Blue
5	White-Blue	White-Blue
6	Orange	Green
7	White-Brown	White-Brown
8	Brown	Brown

Simple Explanation

- T568A: One standard colour order for Ethernet cables.
- T568B: Another standard colour order for Ethernet cables.
- Both standards use 8 wires.
- T568B is commonly used in Ethernet cable connections.

2.Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.
ANS:

Comparison of Ethernet Standards

Ethernet Standard	Speed	Cable Type	Maximum Length	Connector Type
10BASE-T	10 Mbps	Cat3 or higher	100 metres	RJ-45
100BASE-TX	100 Mbps	Cat5 or higher	100 metres	RJ-45
1000BASE-T	1000 Mbps (1 Gbps)	Cat5e or higher	100 metres	RJ-45

Simple Explanation

1. 10BASE-T: Provides 10 Mbps speed.
2. 100BASE-TX: Provides 100 Mbps speed.
3. 1000BASE-T: Provides 1000 Mbps speed (1 Gbps).

3.Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.
ANS:
Connecting Two Laptops Using Ethernet

Answer:

1. I would use a Crossover Cable to connect two laptops directly.
2. It connects the sending wires of one laptop to the receiving wires of the other laptop.
3. It helps both laptops share data directly without a router or switch.

4.List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.
ANS:

Devices That Use Ethernet Connections

No.	Device	Ethernet Standard
1	Wi-Fi Router	1000BASE-T
2	Desktop Computer	1000BASE-T
3	Network Printer	100BASE-TX

Simple Explanation

1. Wi-Fi Router: Provides internet to connected devices.
2. Desktop Computer: Uses Ethernet for a fast internet connection.
3. Network Printer: Uses Ethernet to connect with computers and print documents.